# EAST WEST UNIVERSITY
## Department of Computer Science and Engineering
## B.Sc. in Computer Science and Engineering Program
## Mock Midterm Examination

**Course:** CSE251 Electronic Circuits, Section-X
**Instructor:** Rashedul Amin Tuhin, Senior Lecturer, CSE Department
**Full Marks:** 60 (30 will be counted for final grading)
**Duration:** 80 minutes

**Notes:** There are **SIX** questions, answer ALL of them. Course Outcome (CO), Cognitive Level, Mark, for each question are mentioned at the right margin.

**In the actual exam, the formulas will not be given.**

**Q1.** From the given i-v characteristics graph, **calculate** the following terms: **[CO1, C2, Mark: 20]**

- **a)** the internal resistance of the diode ($r_D$) if the series resistance is given as 3kΩ. [mark:04]
- **b)** the thermal voltage ($V_T$) at 27°C. [mark:04]

$$V_T = \frac{K_B T}{q}$$

- **c)** the Scaling Current ($I_S$). [mark:04]
- **d)** the diode parameter (**n**). [mark:04]

$$I_D = I_S\, e^{\left(\frac{V_D}{nV_T}\right)}$$

- **e)** Explain the PWL model equivalent circuit of a diode. [mark: 04]

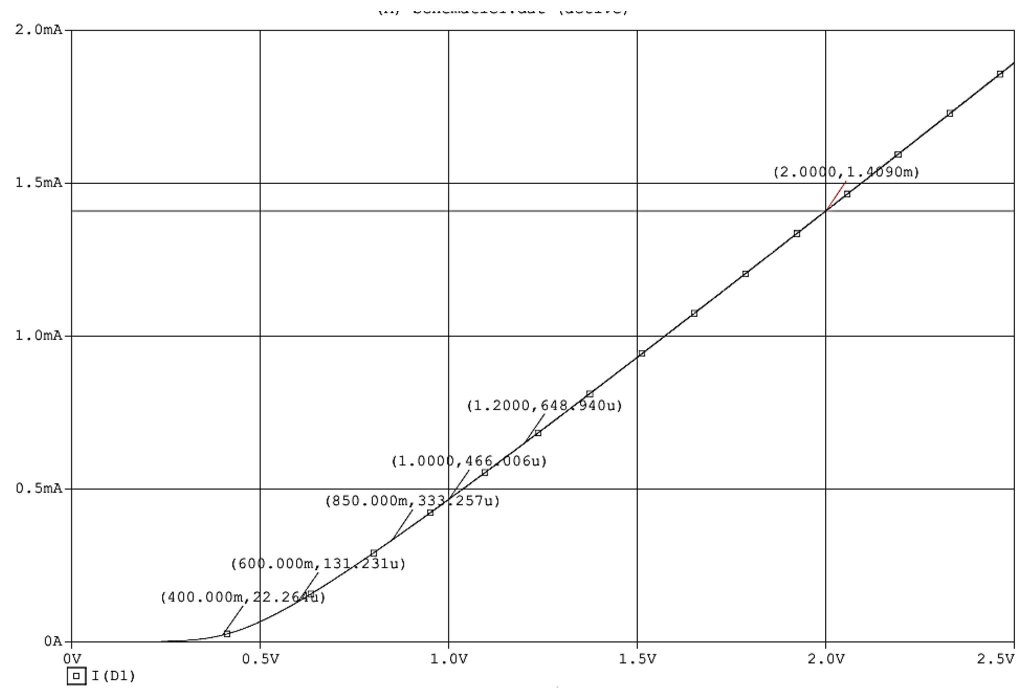



***Figure 1***

**Q2.** Assuming the diodes to be ideal, find the values of I and V in the circuits given in Figure 2. **[CO1, C3, Mark: 12]**

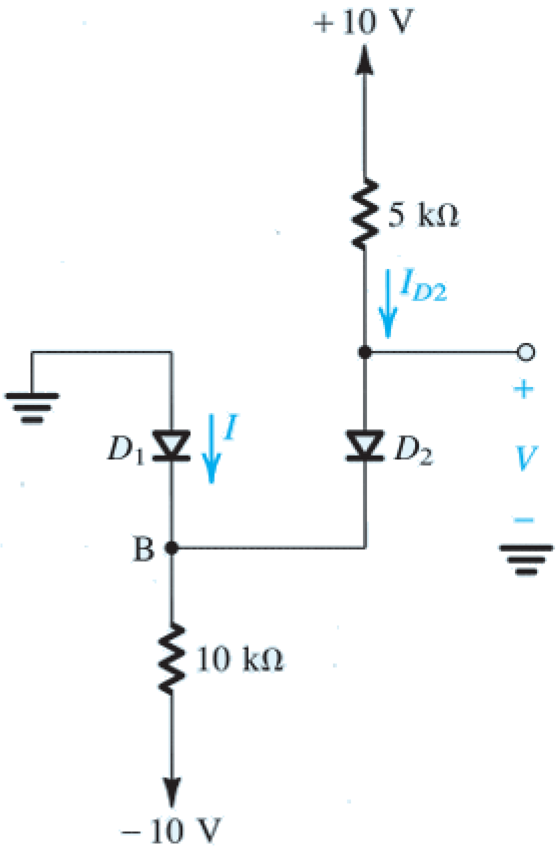
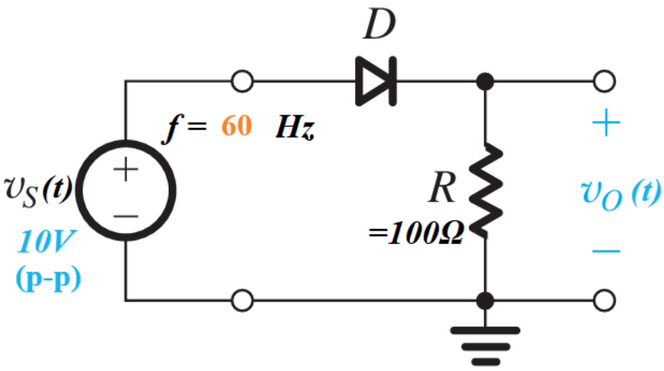
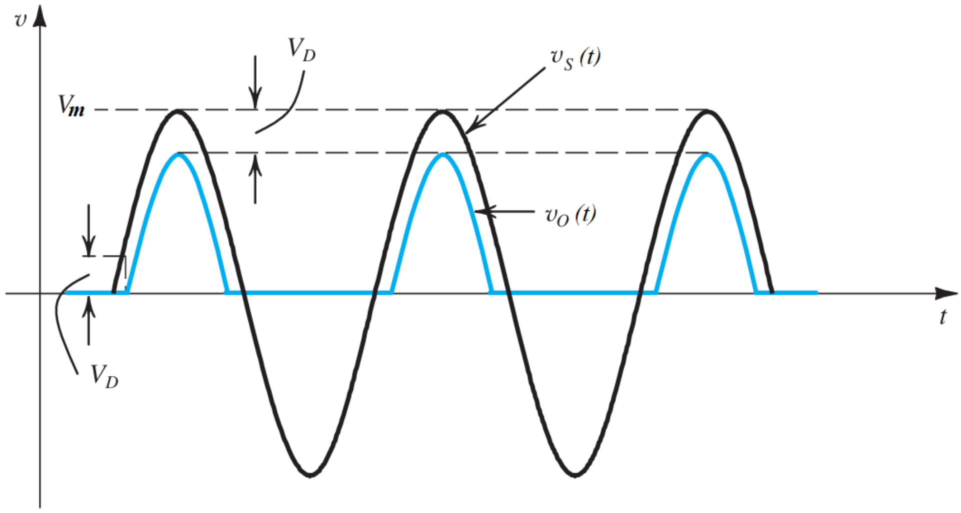



Figure 2

**Q3.** **[CO2, C3, Mark: 12]**






Given,

$$v_s(t) = V_m \sin \omega t = V_m \sin 2\pi f t$$
$$v_S(\Delta t) = V_m \sin \omega \Delta t = V_D = v_S(\theta) = V_m \sin \theta$$
$$\sin \omega \Delta t = \frac{V_D}{V_m} = \sin \theta$$

Consider a simple half wave rectifier with a silicon diode,

- **a)** **sketch** the output waveforms in the time domain ($t$) and **mark** the diode conduction time. [mark:02]
- **b)** **sketch** the output waveforms in the frequency domain ($\theta$) and **mark** the diode conduction angle. [mark:02]
- **c)** **Show that the average** value of the output voltage in a silicon diode half wave rectifier is $(\frac{V_m}{\pi} - \frac{V_D}{2})$, where the symbols carry their usual meaning.

**Calculate** the following terms for the half wave rectifier:

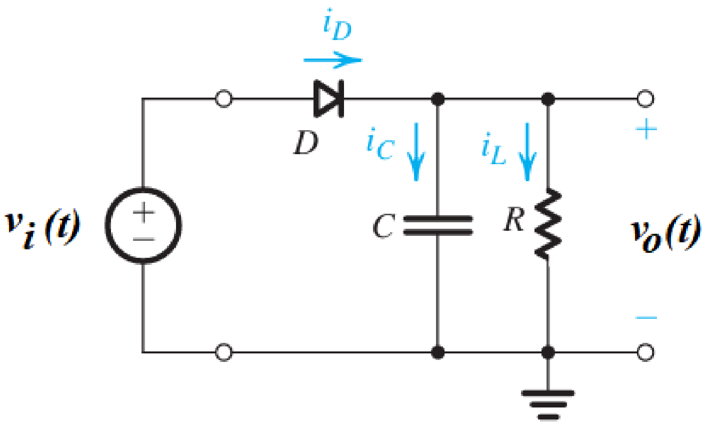
- **d)** *Diode conduction time*, $(\frac{T}{2} - 2\Delta t)$ [mark:02]
- **e)** *Diode conduction angle*, $(\pi - 2\theta)$ [mark:02]
- **f)** Relate among *Diode conduction time* and *Diode conduction angle*. [mark:02]
- **g)** Diode conduction percentage. [mark:02]
- **h)** **Show** how to use *Diode conduction percentage* to calculate *Diode conduction angle* from *Diode conduction time* and vice-versa.
- **i)** *Peak Inverse Voltage*, ($V_m$) [mark:02]
- **j)** *Peak diode current*, $(\frac{V_m - V_D}{R})$ [mark:02]
- **k)** *DC Average value* of the output voltage, $(\frac{V_m}{\pi} - \frac{V_D}{2})$ [mark:02]

**Q2 ***  Repeat the same problem given in Q2 for a silicon diode full wave bridge rectifier. Draw the circuit diagram.

- **a)** **sketch** the input and output waveforms in the time domain ($t$) and **mark** the diode conduction time. [mark:02]

- **b)** **sketch** the input and output waveforms in the frequency domain ($\theta$) and **mark** the diode conduction angle. [mark:02]
- **c)** *Diode conduction time*, ($\frac{T}{2} - 2\Delta t$) [mark:02]
- **d)** *Diode conduction angle*, ($\pi - 2\theta$) [mark:02]
- **e)** *Diode conduction percentage* [mark:02]
- **f)** *Full cycle diode conduction time*, ($T - 4\Delta t$) [mark:02]
- **g)** *Full cycle Diode conduction angle*, ($2\pi - 4\theta$) [mark:02]
- **h)** *Full cycle conduction percentage* ($\frac{T-4\Delta t}{T} \times 100\%$) [mark:02]
- **i)** *Peak Inverse Voltage*, ($V_m - V_D$) [mark:02]
- **j)** *Peak diode current*, ($\frac{V_m - 2V_D}{R}$) [mark:02]
- **k)** *DC Average value* of the output voltage, ($\frac{2V_m}{\pi} - 2V_D$) [mark:02]

**Q3.** **[CO2, C3, Mark: 06]**




Given that, a peak rectifier circuit with the following parameters,

$\mathbf{V_D} = 0.7V$, $\mathbf{V_i} = 70.7V$ (RMS), $\mathbf{V_r} = 5V$, $\mathbf{R} = 100\Omega$, $\mathbf{f} = 60Hz$.

**Derive** the formula to relate between the Ripple voltage (Vr) and the Peak output voltage (Vm).

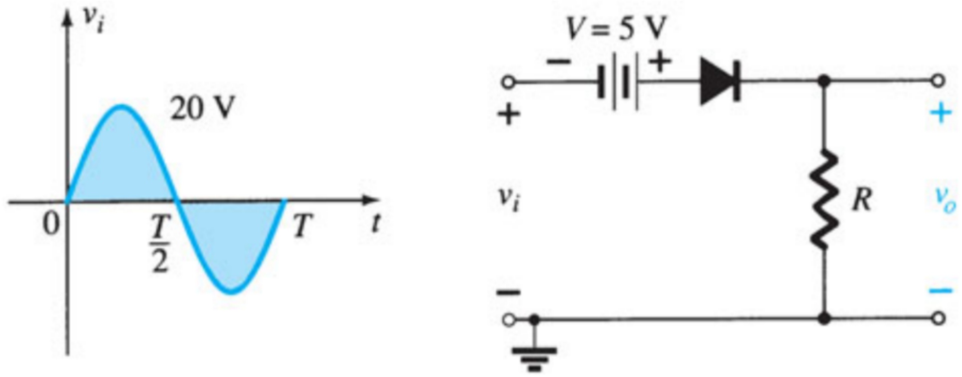
$$V_r = \frac{V_m}{fCR}$$

**Show** that the diode conduction angle, $\theta = \omega\Delta t = \sqrt{\frac{2V_r}{V_m}}$ when $V_r \ll V_m$.

**Calculate** the following terms:

- **a)** the value of the capacitor for the peak rectifier. [mark:02]
- **b)** the diode conduction angle. [mark:02]
- **c)** the peak value of the load current. [mark:02]
- **d)** the average value of the load current. [mark:02]

**Q4.** The input signal, $v_i(t)$ to the circuit is defined as shown in Figure 1. **[CO2, C3, Mark: 06]**




**Determine** the transition voltage for the clipper circuit.
**Sketch** the output waveform, $v_o(t)$. Assume the diode to be an ideal diode.

**Q5.** The input signal, $v_i(t)$ and the circuit are shown in Figure 2. **[CO2, C3, Mark: 06]**
**Sketch** the output waveform, $v_o(t)$. Assume the diode to be a silicon diode.